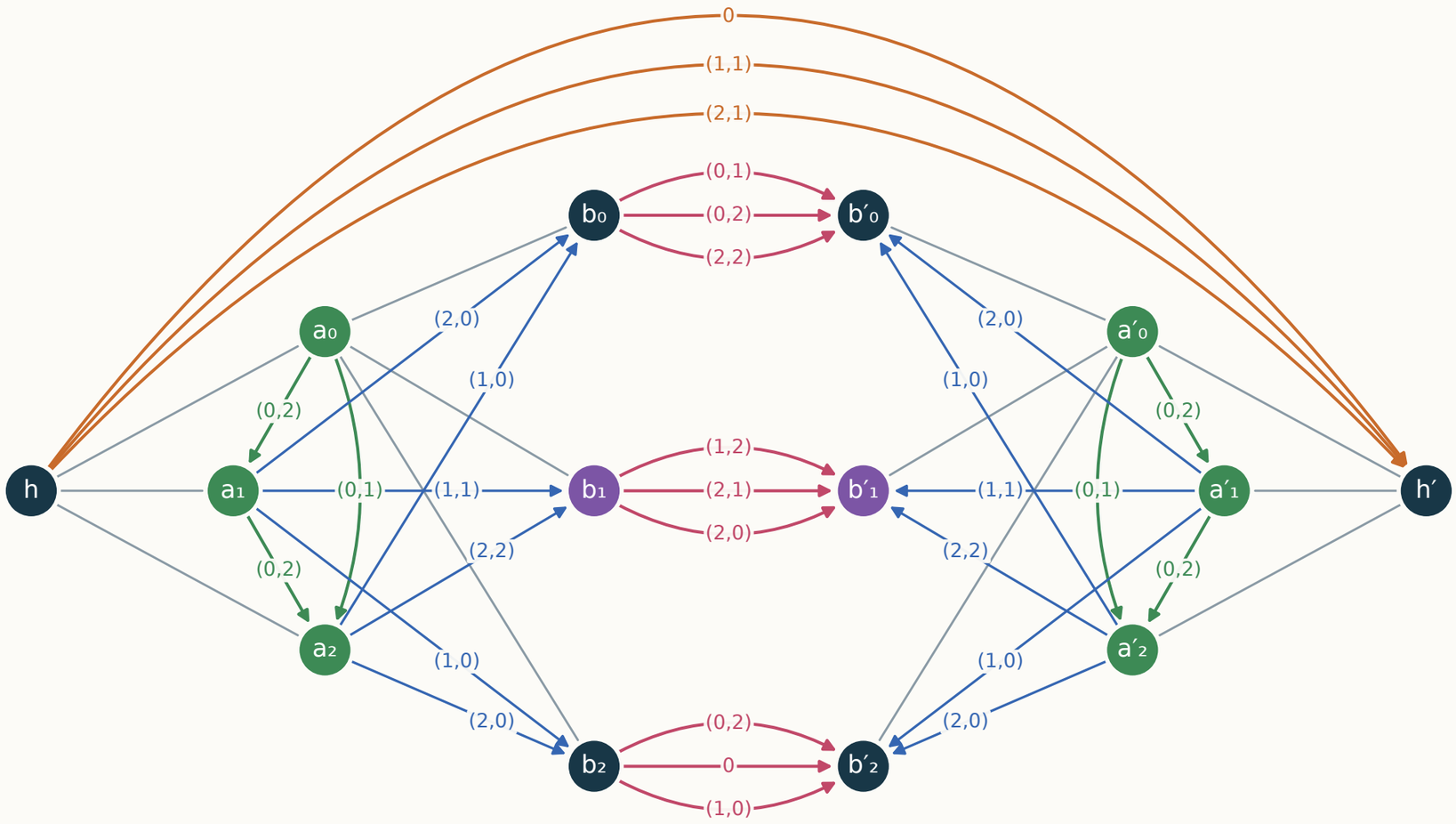
The $Z_3 \times Z_3$ voltage graph behind the 126 vertices

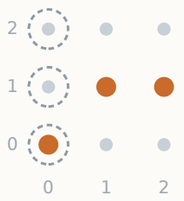
Node u stands for the fibre $\{(u,x) : x \in \mathbb{Z}_3^2\}$. An arrow $u \to v$ with label g means $(u,x) \sim (v, x+g)$; grey edges have voltage 0.



Node colours: the orbits of Aut(G)

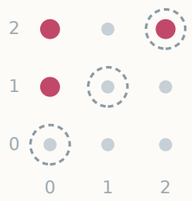
- h, h′, b₀, b₂, b′₀, b′₂ (54 vertices)
- triangle fibres (54)
- b₁, b′₁ (18)

h → h′



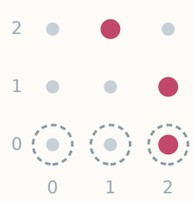
avoids ⟨(0,1)⟩

b₀ → b′₀



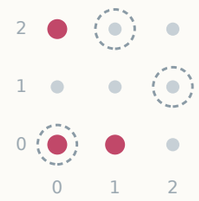
avoids ⟨(1,1)⟩

b₁ → b′₁



avoids ⟨(1,0)⟩

b₂ → b′₂



avoids ⟨(1,2)⟩

Each triple of parallel voltages is a (3,3,3,1) relative difference set in AG(2,3): its six differences are distinct and miss one line through 0 (dashed rings). The four directions of AG(2,3) each occur once.

Triangles $a_l \to a_m$ carry $(m-l)(0,2)$; edges $a_l \to b_j$ carry $l\,w_j$ with $w = (2,0), (1,1), (1,0)$. The right half repeats the voltages of the left half.